

Wazuh + VirusTotal Integration on GCP

Complete Step-by-Step Guide for Beginners

File Integrity Monitoring • Threat Intelligence • Automated Response

*Author: Samson Ejim
Cybersecurity Enthusiast*

Google Cloud Platform | Ubuntu 22.04 LTS | Wazuh 4.7.5

■ Table of Contents

Introduction & Prerequisites	3
Architecture Overview	3
Detection Pipeline	4
Phase 1 — GCP Infrastructure Setup	4
Phase 2 — Install Wazuh Server	5
Phase 3 — Install Wazuh Agent on Ubuntu Endpoint	6
Phase 4 — Configure File Integrity Monitoring (FIM)	7
Phase 5 — VirusTotal Integration	8
Phase 6 — Automated Malware Response (Active Response)	9
Phase 7 — End-to-End Validation Test	10
Troubleshooting Common Errors	11
Summary & Next Steps	12

■ Introduction

This guide walks you through building a complete Security Operations Center (SOC) lab from scratch on Google Cloud Platform (GCP). You will deploy Wazuh — an open-source SIEM platform — configure it to monitor files in real-time, connect it to VirusTotal for threat intelligence, and set up automated malware removal. By the end, any malicious file dropped on your monitored server will be **automatically detected and deleted within 60 seconds** — with no human intervention required.

What You Will Need

Google Account	A free Google account to access Google Cloud Platform
GCP Free Trial	GCP offers \$300 free credits for new accounts — enough for this lab
VirusTotal Account	Free account at virustotal.com for the API key
Basic Linux knowledge	Ability to run commands in a terminal (SSH)
Web Browser	To access the Wazuh dashboard and GCP Console

■ TIP

This guide is designed for complete beginners. Every command is explained. Take it one step at a time!

■ Architecture Overview

The lab uses two virtual machines (VMs) on GCP. The **Wazuh Server** is the brain — it collects alerts, runs the dashboard, and calls VirusTotal. The **Ubuntu Endpoint** is the monitored machine — it runs the Wazuh Agent which watches files and executes automated responses.

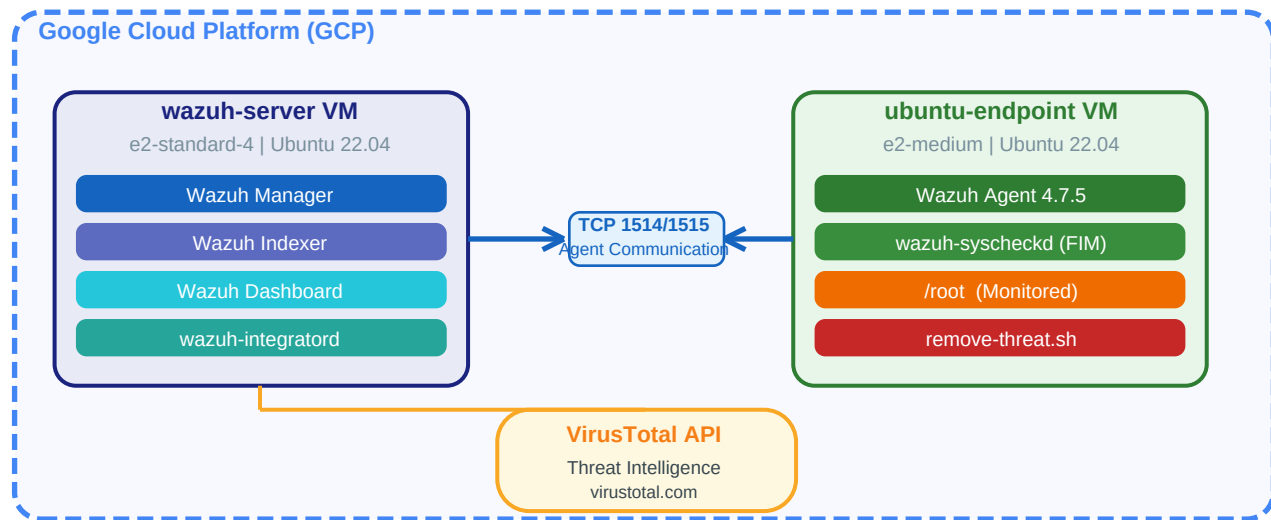


Figure 1: GCP Infrastructure — Wazuh Server communicates with Ubuntu Endpoint via Wazuh Agent

■ Detection Pipeline

When a file is dropped in the monitored **/root** directory, a chain of automated actions fires instantly:

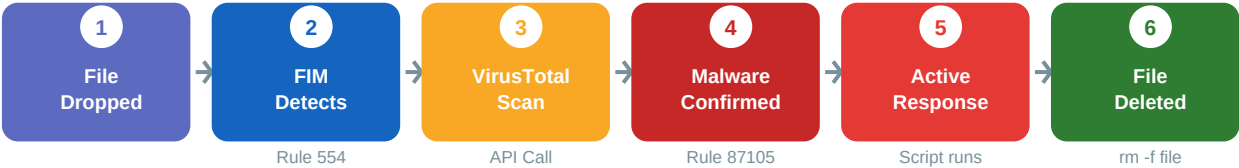


Figure 2: Full Detection & Response Pipeline

PHASE 1 GCP Infrastructure Setup

Before installing any software, you need to set up your cloud environment. This involves creating a project and launching two virtual machines.

1 Create a GCP Project

1. Open your browser and go to **console.cloud.google.com**
2. Click the project dropdown at the top → **New Project**
3. Name it **wazuh-soc-lab** and click **Create**
4. Make sure the new project is selected in the dropdown

2 Create the Wazuh Server VM

Go to **Compute Engine** → **VM Instances** → **Create Instance**

Name	wazuh-server
Region	us-central1 (or any region close to you)
Machine type	e2-standard-4 (4 vCPU, 16 GB RAM) ← Wazuh requires this minimum
Boot disk OS	Ubuntu 22.04 LTS
Boot disk size	50 GB SSD
Firewall	Check both: Allow HTTP traffic AND Allow HTTPS traffic

■ ■ WARNING

The e2-standard-4 machine type is important. Wazuh will fail to start on smaller machines.

3 Create the Ubuntu Endpoint VM

Create a second VM with these settings:

Name	ubuntu-endpoint
Machine type	e2-medium (2 vCPU, 4 GB RAM)
Boot disk OS	Ubuntu 22.04 LTS
Boot disk size	20 GB SSD

4 Configure GCP Firewall Rules

Go to **VPC Network** → **Firewall** → **Create Firewall Rule** and create a rule named **wazuh-ports** with these settings:

Direction	Ingress
Targets	All instances in the network
Source IP ranges	0.0.0.0/0
TCP Ports to open	1514, 1515, 443, 55000

■ ■ NOTE

Port 1514/1515 = Wazuh agent communication. Port 443 = Dashboard. Port 55000 = Wazuh API.

PHASE 2 Install Wazuh Server (All-in-One)

The Wazuh all-in-one installer sets up the Manager, Indexer (database), and Dashboard in a single automated script. This takes about 10–15 minutes.

1 SSH into your wazuh-server VM

In GCP Console, find **wazuh-server** in the VM list. Click the **SSH** button on the right. A terminal window will open in your browser.

2 Download the installer files

```
curl -sO https://packages.wazuh.com/4.7/wazuh-install.sh
curl -sO https://packages.wazuh.com/4.7/config.yml
```

3 Edit the config file with your server's internal IP

First, find your server's internal IP: in GCP Console, click on **wazuh-server** and look for **Internal IP** (e.g. 10.128.0.4). Then edit the config:

```
nano config.yml
```

Replace all occurrences of **<indexer-node-ip>**, **<wazuh-manager-ip>**, and **<dashboard-ip>** with your actual internal IP address. Save with **Ctrl+X → Y → Enter**.

4 Run the installer

```
sudo bash wazuh-install.sh -a
```

■ ■ WARNING

This takes 10-15 minutes. At the end it prints your admin username and password. **SAVE THESE** — you need them to log in!

5 Access the Wazuh Dashboard

Open a new browser tab and go to:

```
https://<YOUR-WAZUH-SERVER-EXTERNAL-IP>
```

Your browser may warn about an insecure certificate — click **Advanced** → **Proceed**. Login with the admin credentials printed by the installer.

PHASE 3 Install Wazuh Agent on Ubuntu Endpoint

The Wazuh Agent is installed on the machine you want to monitor (ubuntu-endpoint). It reports events back to the Wazuh Server. **Important:** The agent version must match the server version (4.7.5).

1 SSH into ubuntu-endpoint VM

In GCP Console, find **ubuntu-endpoint** in the VM list and click **SSH**.

2 Import the Wazuh GPG security key

This key verifies that the software you download is genuine Wazuh software.

```
sudo bash -c 'curl -s https://packages.wazuh.com/key/GPG-KEY-WAZUH | gpg \
--no-default-keyring \
--keyring gnupg-ring:/usr/share/keyrings/wazuh.gpg --import'

sudo chmod 644 /usr/share/keyrings/wazuh.gpg
```

■ ■ WARNING

Use `sudo bash -c '...'` to wrap the whole command. Running `sudo` only on `curl` causes a permission error.

3 Add the Wazuh software repository

```
echo "deb [signed-by=/usr/share/keyrings/wazuh.gpg] \
https://packages.wazuh.com/4.x/apt/ stable main" | \
sudo tee /etc/apt/sources.list.d/wazuh.list

sudo apt-get update
```

4 Install Wazuh Agent version 4.7.5

Download and install the specific version that matches your server:

```
wget https://packages.wazuh.com/4.x/apt/pool/main/w/wazuh-agent/wazuh-agent_4.7.5-1_amd64.deb

sudo WAZUH_MANAGER="<WAZUH_SERVER_INTERNAL_IP>" dpkg -i
wazuh-agent_4.7.5-1_amd64.deb
```

■ ■ NOTE

Replace with your wazuh-server internal IP (e.g. 10.128.0.4). Find it in GCP Console.

5 Start the Wazuh Agent

```
sudo systemctl daemon-reload
sudo systemctl enable wazuh-agent
sudo systemctl start wazuh-agent
sudo systemctl status wazuh-agent
```

You should see **Active: active (running)** in green. If you see an error about 'Invalid server address', edit the config file:

```
sudo nano /var/ossec/etc/ossec.conf
# Find the <client> block and set your manager IP:
# <address>10.128.0.4</address> ← your wazuh-server internal IP
# Then restart: sudo systemctl restart wazuh-agent
```

■ TIP

Check the Wazuh dashboard — you should now see '1 Active agent' on the home page!

PHASE 4 Configure File Integrity Monitoring (FIM)

FIM watches directories for any changes — files being added, modified, or deleted. We will configure it to watch the **/root** directory in real-time. This is done on the **ubuntu-endpoint**.

1 Edit the agent configuration file

```
sudo nano /var/ossec/etc/ossec.conf
```

Use **Ctrl+W** (search) and type **syscheck** to jump to the right section.

2 Modify the syscheck block

Make these three changes inside the existing `<syscheck>` block:

Change	From	To
Frequency	<code><frequency>43200</frequency></code>	<code><frequency>300</frequency></code>
Add /root monitoring	<code>(not present)</code>	<code><directories realtime="yes" check_all="yes" report_changes="yes">/root</directories></code>
Update /etc line	<code><directories>/etc, ...</directories></code>	<code><directories check_all="yes">/etc, ...</directories></code>

3 Save and restart the agent

```
# Save: Ctrl+X → Y → Enter
sudo systemctl restart wazuh-agent
```

4 Test that FIM is working

```
sudo touch /root/testfile.txt
```

Wait 30 seconds, then check the Wazuh dashboard: **Modules** → **Integrity Monitoring**. You should see an alert showing **/root/testfile.txt** was added (Rule 554).

PHASE 5 VirusTotal Integration

VirusTotal is a free online service that checks files and URLs against 90+ antivirus engines. We will connect Wazuh to VirusTotal so every FIM alert automatically gets checked for malware. This is done on the **wazuh-server**.

1 Get your free VirusTotal API Key

1. Go to **virustotal.com** and create a free account
2. Click your profile icon (top right) → **API Key**
3. Copy your API key — you will need it in the next step

NOTE

The free VirusTotal API allows 4 requests per minute — more than enough for a lab environment.

2 Edit the Wazuh server config

SSH into **wazuh-server** and open the config file:

```
sudo nano /var/ossec/etc/ossec.conf
```

Press **Ctrl+End** to jump to the very bottom. Add the following block just **above** the last **</ossec_config>** tag:

```
<integration>
<name>virustotal</name>
<api_key>YOUR_VIRUSTOTAL_API_KEY_HERE</api_key>
<rule_id>554,550,553,552,551</rule_id>
<alert_format>json</alert_format>
</integration>
```

WARNING

Replace **YOUR_VIRUSTOTAL_API_KEY_HERE** with your actual API key. Keep the angle brackets and tags exactly as shown.

3 Restart Wazuh Manager

```
sudo systemctl restart wazuh-manager
sudo systemctl status wazuh-manager
```

Look for **wazuh-integratord** in the process list — this confirms the VirusTotal integration is active.

4 Verify VirusTotal is being called

Watch the integration log on **wazuh-server**:

```
sudo tail -f /var/ossec/logs/integrations.log
```

Then on ubuntu-endpoint, create a test file:

```
sudo touch /root/testfile2.txt
```

You should see activity in the log — a file hash being sent to VirusTotal. On the dashboard under **Security Events**, you will see Rule **87104** (No positives found) for clean test files.

PHASE 6 Automated Malware Response (Active Response)

Active Response tells Wazuh to automatically run a script when a specific alert fires. We will create a script that deletes any file confirmed as malware by VirusTotal (Rule 87105).

1 Create the removal script on ubuntu-endpoint

```
sudo nano /var/ossec/active-response/bin/remove-threat.sh
```

Paste this script exactly:

```
#!/bin/bash
# Wazuh Active Response: Remove confirmed malware

LOCAL=$(dirname $0)
cd $LOCAL
cd ../
PWD=$(pwd)

read INPUT_JSON

FILENAME=$(echo $INPUT_JSON | python3 -c \
'import sys, json; \
data = json.load(sys.stdin); \
print(data["parameters"]["alert"]["data"]["virustotal"]["source"]["file"])')

echo "$(date) - Removing malware: $FILENAME" >>
/var/ossec/logs/active-responses.log
rm -f "$FILENAME"
echo "$(date) - Successfully removed: $FILENAME" >>
/var/ossec/logs/active-responses.log
```

Save, then set the correct permissions:

```
sudo chmod 750 /var/ossec/active-response/bin/remove-threat.sh
sudo chown root:wazuh /var/ossec/active-response/bin/remove-threat.sh
```

2 Register the command on wazuh-server

SSH into **wazuh-server** and edit the config:

```
sudo nano /var/ossec/etc/ossec.conf
```

Find the existing **<command>** blocks (search with Ctrl+W for 'command'). Add your new command block right after the last **</command>** tag:

```
<command>
<name>remove-threat</name>
<executable>remove-threat.sh</executable>
<timeout_allowed>no</timeout_allowed>
</command>
```

Then find the commented-out **<active-response>** section and add your active response block right after the closing -->:

```
<active-response>
<disabled>no</disabled>
<command>remove-threat</command>
<location>local</location>
<rules_id>87105</rules_id>
</active-response>
```

■ NOTE

Rule 87105 is the VirusTotal 'malware confirmed' rule. This active response only fires when malware is CONFIRMED — not for clean files.

3 Restart Wazuh Manager

```
sudo systemctl restart wazuh-manager
```

PHASE 7 End-to-End Validation Test

Now test the complete pipeline using the **EICAR test file** — an industry-standard, completely harmless file that every antivirus treats as malware for testing purposes. It contains no actual malicious code.

■ TIP

EICAR is 100% safe. It is a 68-byte text file that antivirus engines flag by design for testing purposes.

1 Drop the EICAR test file on ubuntu-endpoint

```
sudo curl -Lo /root/eicar.com https://secure.eicar.org/eicar.com.txt
```

2 Wait 60 seconds then verify the file is gone

```
ls /root/eicar.com

# Expected result:
# ls: cannot access '/root/eicar.com': No such file or directory
```

■ TIP

If the file was automatically deleted, the active response worked! The full pipeline is complete.

3 Check the Active Response log

```
sudo cat /var/ossec/logs/active-responses.log

# You should see:
# Mon Apr 27 17:01:38 UTC 2026 - Removing malware: /root/eicar.com
# Mon Apr 27 17:01:38 UTC 2026 - Successfully removed: /root/eicar.com
```

4 Verify on the Wazuh Dashboard

Go to **Modules** → **Security Events**. You should see this full alert chain:

Rule ID	Level	Description	What it means
554	5	File added to the system	FIM detected eicar.com in /root
87104	3	No positives found	Previous clean test files
87105	12	66 engines detected this file	MALWARE CONFIRMED by VirusTotal

■ Troubleshooting Common Errors

Error 1: GPG Permission Denied error

Symptom: gpg: failed to create temporary file '/usr/share/keyrings/.#lk...': Permission denied

Fix: Use sudo bash -c '...' to wrap the entire curl | gpg pipeline so both commands run as root.

```
sudo bash -c 'curl -s https://packages.wazuh.com/key/GPG-KEY-WAZUH | gpg \
--no-default-keyring --keyring gnupg-ring:/usr/share/keyrings/wazuh.gpg --import'
```

Error 2: Agent fails to start: Invalid server address

Symptom: wazuh-agentd: ERROR: (4112): Invalid server address

Fix: The agent config does not have the manager IP set. Edit ossec.conf and set it manually.

```
sudo nano /var/ossec/etc/ossec.conf
# Find and set your manager IP
# Then: sudo systemctl restart wazuh-agent
```

Error 3: Agent shows 0 agents on dashboard (version mismatch)

Symptom: Server: 4.7.5 but agent installed: 4.14.x — they cannot communicate

Fix: Reinstall the agent at the exact same version as the server (4.7.5).

```
sudo apt-get remove --purge wazuh-agent -y && sudo rm -rf /var/ossec
wget https://packages.wazuh.com/4.x/apt/pool/main/w/wazuh-agent/wazuh-agent_4.7.5-1_amd64.deb
sudo WAZUH_MANAGER="" dpkg -i wazuh-agent_4.7.5-1_amd64.deb
```

Error 4: eicar.com not deleted after test

Symptom: File still exists after 60 seconds — active response did not fire

Fix: Check that remove-threat.sh has correct permissions and the active-response block uses rules_id 87105.

```
sudo chmod 750 /var/ossec/active-response/bin/remove-threat.sh
sudo chown root:wazuh /var/ossec/active-response/bin/remove-threat.sh
sudo systemctl restart wazuh-manager
```

■ Summary & What You Accomplished

Phase	Component	What was achieved
Phase 1	GCP Infrastructure	Created 2 VMs (wazuh-server + ubuntu-endpoint) with correct firewall rules
Phase 2	Wazuh Server	Deployed full Wazuh stack (Manager + Indexer + Dashboard) using all-in-one installer
Phase 3	Wazuh Agent	Installed matching agent v4.7.5 on endpoint, resolved version mismatch
Phase 4	File Integrity Monitoring	Configured real-time FIM on /root directory — detects file changes in <30 seconds
Phase 5	VirusTotal Integration	Connected Wazuh to VirusTotal API for automatic threat intelligence enrichment
Phase 6	Active Response	Built remove-threat.sh script that auto-deletes malware when Rule 87105 fires
Phase 7	Validation	EICAR test file flagged by 66/90 engines, automatically deleted in <60 seconds

■ Key Skills Demonstrated

- Cloud infrastructure deployment on Google Cloud Platform (GCP)
- SIEM deployment and configuration (Wazuh)
- Host-based intrusion detection using File Integrity Monitoring (FIM)
- Third-party API integration for threat intelligence (VirusTotal)
- Automated incident response scripting (Bash)
- Linux system administration and troubleshooting
- Security alert triage and interpretation
- End-to-end SOC pipeline design and validation

■■ Suggested Next Projects

Suricata IDS/IPS	Network-based intrusion detection — catch attacks before they reach the host
Splunk SIEM Lab	Industry-leading SIEM used in most enterprise SOC environments
Velociraptor DFIR	Digital forensics and incident response on compromised endpoints
Honeypot Deployment	Attract and study real attacker behaviour in a safe environment

Built with dedication by **Samson Ejim** — Cybersecurity Enthusiast
github.com/samsonejim/wazuh-virustotal-soc-lab